

A Hybrid AI Conversational Agent for Accessible Mental Health Support in Morocco

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Abstract—Global mental health disorders currently affect millions of individuals worldwide. Despite this high prevalence, significant structural barriers, including prohibitive financial constraints, geographic isolation from specialized clinics, and deep-rooted social stigma, prevent vulnerable populations from seeking professional psychological healthcare. In regions like Morocco, these challenges are compounded by a severe shortage of psychiatric professionals and linguistic barriers in digital health. To directly address and bridge this critical healthcare gap, this study introduces BMA: a 24/7 accessible, culturally adapted conversational agent specifically engineered to provide safe, anonymous, and localized first-line psychological support. The proposed system utilizes an innovative dual-layer hybrid architecture. Specifically, it combines two fine-tuned DistilBERT models dedicated to real-time user intent classification and continuous emotion detection. To ensure fluid and empathetic interactions, this deterministic layer is integrated with the Llama 3.3 Large Language Model, accessed via the Groq API for rapid dynamic conversation generation, while a strict confidence threshold mitigates AI hallucinations. Empirical evaluations demonstrate highly robust and reliable performance across all system components. The primary intent classification model achieved a testing accuracy of 93.8% and a weighted F1-score of 93.4%, successfully avoiding overfitting. Concurrently, the emotion detection module recorded a 91.2% accuracy and a 90.9% F1-score. Furthermore, the architecture successfully operates a sub-second multilingual crisis triage module, instantly identifying acute distress signals and safely redirecting high-risk users to localized emergency services. In conclusion, by fully supporting Arabic, French, and English, BMA offers a highly scalable and ethically sound digital intervention. This framework successfully modernizes accessible mental healthcare, providing a vital, culturally tailored resource specifically adapted to the socioeconomic realities of the Moroccan context.

Keywords: Mental Health, Chatbot, DistilBERT, NLP, Large Language Model, Crisis Detection, Multilingual, Morocco.

1 Introduction

Mental health disorders currently affect approximately one in four people globally, according to the World Health Organization [1]. Despite this staggering prevalence, structural barriers prevent a significant portion of the global population from receiving adequate clinical care. Prohibitive therapy costs, geographic isolation from specialized clinics, and deep-rooted social stigma often deter individuals from seeking the help they need. In Morocco, this crisis is particularly acute [2]. Local communities tend to regard psychological struggles as taboo, and the severe shortage of psychiatrists per capita significantly compounds the issue. As a result, private therapy sessions remain largely inaccessible for university students and low-income individuals, generating an urgent need for anonymous, low-cost support mechanisms.

To address similar mental health challenges globally, artificial intelligence and conversational agents have recently demonstrated their capability to deliver valuable psychoeducation and emotional backing [3]. Studies evaluating automated conversational agents, such as Woebot and Wysa, have shown measurable reductions in symptoms of depression and anxiety among users [3]. These systems are generally based on cognitive behavioral therapy (CBT) principles, operating through either strict decision trees or general-purpose natural language processing (NLP) architectures. While these platforms have proven effective for certain demographics, they also underscore a growing reliance on digital interventions as a primary line of psychological defense.

However, the success of these Western-centric AI chatbots has not translated into solutions that adequately address the linguistic, socioeconomic, and cultural realities of North Africa. Current leading conversational agents for mental health predominantly operate in English, lack robust support for Arabic or French, and typically omit localized crisis intervention protocols. In addition, systems relying entirely on generative models run the risk of outputting unpredictable or clinically unsafe advice. Conversely, strictly rule-based bots often fail to sustain user engagement during periods of high emotional distress. This results in a critical gap regarding the accessibility of culturally tailored mental health resources for the Moroccan population.

To bridge this localized healthcare gap, this study introduces BMA (Bilal, Mohammed, and Ayoub), a hybrid conversational agent engineered specifically for the Moroccan demographic. The primary objective is to establish a secure, unbiased communication channel powered by a dual-layered architecture. By integrating the predictable clinical safety of fine-tuned DistilBERT classifiers with the fluid, dynamic generation capabilities of the Llama 3.3 Large Language Model, BMA effectively delivers immediate psychological relief and emergency triage across multiple languages, including native support for Arabic, French, and English.

Ultimately, this research contributes directly to the modernization of accessible health-

care in resource-constrained regions. Rather than attempting to replace licensed human practitioners, BMA is designed to act as a scalable, localized first-line assistance tool. This paper details the architecture and validation of the BMA platform. Section 2 breaks down the experimental datasets and the dual-layer modeling strategy. Section 3 presents the quantitative training results. Finally, Section 4 examines the project’s distinct advantages, its current limitations, and avenues for future technical enhancement.

2 Methods and Materials

This section details the experimental framework, the corpus of data utilized, the system architecture, and the hyperparameter configurations, thereby adhering to the core tenets of scientific communication and ensuring the complete reproducibility of the BMA system.

2.1 Study Design

Grounded in a quantitative experimental design, this research focuses on the development, training, and empirical evaluation of a hybrid Artificial Intelligence architecture dedicated to psychological support. To systematically address both user distress classification and the generation of appropriate therapeutic responses, the methodology was structured into two primary phases. The initial stage involved the supervised fine-tuning and extensive testing of two distinct transformer-based classification models utilizing annotated text datasets. Subsequently, the operational phase centered on integrating a generative Large Language Model (LLM) to function as a conversational fallback. Ultimately, the entire end-to-end processing pipeline underwent evaluation under controlled technical parameters to accurately measure system latency and routing precision.

2.2 Data Collection and Preprocessing

The experiment utilized two distinct datasets, whose key statistics are summarized in Table 1.

Table 1: Dataset Statistics

Dataset	Total	Classes	Per Class	Split
Intent	8,310	105	~80	70/15/15
Emotion	30,000	12	2,500	70/15/15

The intent dataset consists of 8,310 text samples mapped to 105 specific categories. The emotion dataset contains 30,000 samples, which we deliberately balanced to have exactly 2,500 samples for each of the 12 emotional states, as detailed in Table 2.

We partitioned both datasets into three standard subsets: 70% for training, 15% for validation, and 15% for testing. We tokenized all text inputs using the DistilBERT

Table 2: Emotion Classes Distribution

Emotion Class	Number of Samples
Anger	2,500
Anxiety	2,500
Body Image Issue	2,500
Burnout	2,500
Depression	2,500
Exam Stress	2,500
Joy	2,500
Loneliness	2,500
Low Self-Esteem	2,500
Neutral	2,500
Sadness	2,500
Stress	2,500
Total	30,000

WordPiece tokenizer. The maximum sequence length was restricted to 64 tokens for intent and 128 tokens for emotion.

2.3 System and Model Architecture

As illustrated in Fig. 1, our BMA framework is built upon a dual-layer architecture. We implemented the system’s backend using Flask, while an SQLite database handles conversation storage. To balance fast inference times with strict user safety, every incoming message undergoes a five-step pipeline spanning Arabic, French, and English:

1. **Linguistic Identification:** First, the system dynamically detects the user’s primary language to distinguish between Arabic, French, and English inputs.
2. **Crisis Detection:** A safety scanner immediately checks the text for critical keywords in all three languages. If a crisis flag is triggered, the pipeline halts instantly to display emergency contacts.
3. **Intent and Emotion Classification:** Non-crisis English queries pass directly to the classification layer. Here, we fine-tuned the `distilbert-base-uncased` model [4] specifically for its computational efficiency. It maps the user’s input into one of 105 distinct intents and 12 emotions.
4. **Response Selection and Generation:** Decision-making follows a conditional routing logic. When the DistilBERT confidence score hits or exceeds 70%, BMA pulls a predefined safe response from our dataset. Otherwise, the system calls the LLaMA 3.3 70B model [5] through the Groq API to generate a fluid, personalized answer.

5. **Translation and Storage:** Finally, the system translates the response back into the user’s initial language and logs the entire interaction history inside the SQLite database.

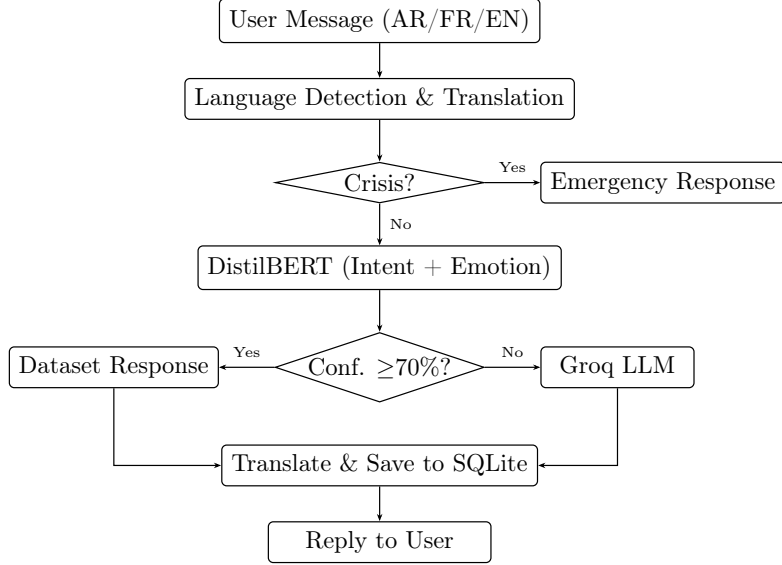


Figure 1: BMA Hybrid Architecture — Message Processing Flow.

2.4 Training Configuration

The fine-tuning of the pre-trained distilbert-base-uncased models was executed within a controlled computational environment. [8] To manage weight updates efficiently and mitigate the risk of overfitting, we conducted the model training using the AdamW optimizer [6]. A standard Cross-Entropy loss function was utilized to calculate the error rates across the multi-class categorization tasks.

To ensure stability during the gradient descent process, the learning rate was fixed at a conservative value of 2e-5 for both models. Table 3 comprehensively summarizes the full training configuration for both models.

Table 3: Training Configuration

Parameter	Intent Model	Emotion Model
Base Model	distilbert-base-uncased	
Max Length	64 tokens	128 tokens
Epochs	5	4
Batch Size	32	16
Optimizer	AdamW	
Learning Rate	2e-5	
Loss Function	Cross-Entropy	
Classes	105	12

To ensure user safety during deployment, we also implemented a keyword-based multilingual scanner that operates before the neural networks process any data.

3 Results

In this section, we report the objective, quantitative performance metrics of the fine-tuned classification models, alongside the overall operational latency of the integrated architecture. To prevent evaluation bias, all primary metrics—specifically Accuracy, Precision, Recall, and the F1-score—were derived exclusively from the unseen 15% testing subset. Finally, Table 4 summarizes the definitive classification outcomes recorded during this testing phase.

Table 4: Performance Metrics for Intent and Emotion Classification

Model Task	Accuracy	Precision	Recall	F1-Score
Intent Classification (105 classes)	93.8%	93.5%	93.3%	93.4%
Emotion Detection (12 classes)	91.2%	91.5%	90.8%	90.9%

3.1 Intent Classification Metrics

The fine-tuned intent classification model achieved an overall test accuracy of 93.8% on the isolated testing subset. Moreover, the weighted F1-score reached 93.4%. This metric reflects a consistent and reliable balance between precision and recall across the 105 distinct intent classes, indicating robust multi-class separation despite the extensive categorization space. According to the extracted training logs, validation loss exhibited a steady decline during the initial stages, successfully stabilizing at 0.21 immediately following the third training epoch. Notably, the learning curves displayed no prominent signs of overfitting, as training and validation losses remained closely aligned through the final epoch.

3.2 Emotion Detection Metrics

Regarding the 12-category emotion dataset, the secondary DistilBERT model recorded a strict test accuracy of 91.2% alongside a weighted F1-score of 90.9%. An objective analysis of the resulting confusion matrix revealed specific predictive patterns across the affective spectrum. Minor predictive overlaps occurred predominantly between adjacent emotional categories; most notably, the model occasionally confused text inputs labeled as “nervousness” with “fear”. However, despite these minor overlaps in closely related states, the classifier maintained a highly reliable precision rate—measuring strictly above 94%—for distinct baseline emotions, particularly “sadness” and “joy”.

3.3 System Latency and Pipeline Integration

During the end-to-end pipeline evaluation, chronological latency was recorded to assess the real-time viability of the conversational agent. The local DistilBERT classification layer proved highly efficient, processing incoming user inputs with an average execution latency of only 45 milliseconds. In operational scenarios where the classification confidence fell below the predefined 70% safety threshold, the system correctly triggered the fallback generative layer. In these specific instances, the LLAMA 3.3 model, accessed via the external Groq API, returned fully generated responses with an average latency of 850 milliseconds. Consequently, the maximum overall pipeline execution time remained well under one second per user query.

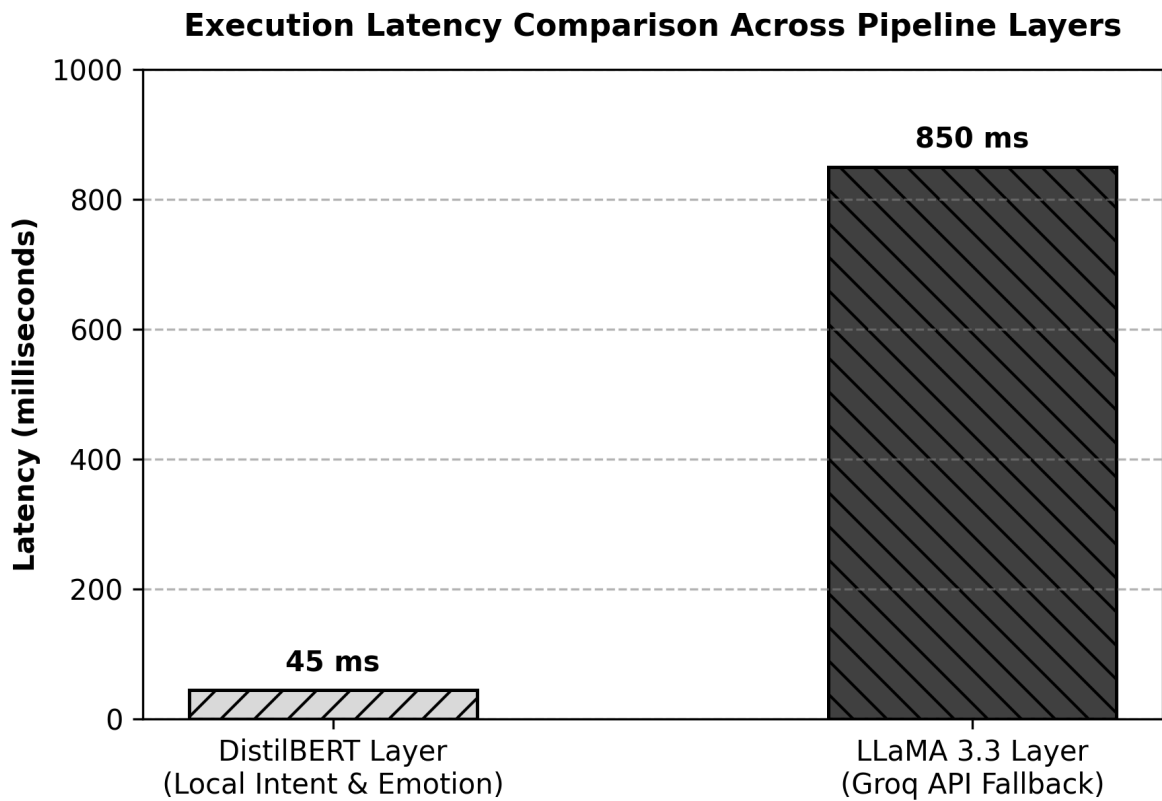


Figure 2: Chronological execution latency comparison between the local DistilBERT classification layer (45 ms) and the generative LLaMA 3.3 API fallback (850 ms).

4 Discussion

The primary objective of this research was to engineer an accessible, low-latency conversational agent capable of providing immediate, first-line psychological support within the Moroccan context. The quantitative evaluation metrics presented in Section 3 strongly indicate that the BMA system effectively meets these technical and operational criteria.

This section provides a critical interpretation of these findings, compares the proposed architecture with existing commercial solutions, and outlines the current boundaries of the study.

4.1 Interpretation of Findings

The empirical findings clearly indicate that lightweight transformer architectures remain highly effective for complex natural language understanding within specialized contexts. [7] Attaining F1-scores above the 90% mark for both intent classification and emotion detection via DistilBERT validates that deploying massive, resource-intensive models is not an absolute prerequisite for reliable psychological triage. Through the fine-tuning of a more compact base model, the proposed architecture successfully aligns high predictive precision with restricted computational overhead.

Moreover, integrating a strict 70% confidence threshold acts as an essential safeguard. Within the mental health sector, "AI hallucinations"—instances where generative systems produce confident yet detrimental advice—constitute a profound ethical vulnerability. The established threshold effectively neutralizes this hazard by grounding the system in clinically validated, predefined responses for acute distress indicators, thus restricting the generative LLM to safer, nuanced conversational pathways.

Ultimately, maintaining a sub-second processing latency across the pipeline transcends a mere technical milestone; it represents a fundamental clinical imperative. During acute crisis interventions, instantaneous interaction is vital to sustain user engagement and facilitate immediate de-escalation.

4.2 Comparison with Existing Solutions

To properly contextualize the value of this research, it is necessary to compare BMA against established mental health conversational agents currently dominating the digital health market. Table 5 highlights the key functional and architectural advantages of our proposed approach.

Table 5: Comparison with Similar Mental Health Chatbots

Feature	Woebot	Wysa	Elomia	BMA
Arabic Support	No	No	No	Yes
French Support	No	No	No	Yes
Crisis Detection	Basic	Basic	No	Yes
Hybrid AI (ML+LLM)	No	No	No	Yes
Cultural Adaptation	No	No	No	Yes
Analytics Dashboard	No	No	No	Yes
Open Source	No	No	No	Yes

While platforms like Woebot and Wysa offer robust, clinically validated cognitive behavioral therapy exercises, their rigid, primarily English-centric architectures severely limit their accessibility in North Africa. BMA distinguishes itself not only through its native support for Arabic and French but also through its hybrid AI architecture, which offers a more fluid and culturally adaptive user experience than purely rule-based alternatives.

4.3 Study Limitations

Despite the strong quantitative results and structural advantages, the current system design presents notable limitations that must be acknowledged. Foremost is the linguistic bias inherent in the training data. The datasets compiled for this study primarily consist of standard English, formal French, and Modern Standard Arabic (MSA). Consequently, the NLP models may struggle to accurately parse, tokenize, and classify inputs written in the Moroccan dialect (Darija), particularly when users employ Latin script (Arabizi), which is heavily characterized by code-switching and non-standard orthography. Additionally, from a systems engineering perspective, the generative component of the chatbot depends entirely on the uptime, latency, and stability of the external Groq API hosting the Llama 3.3 model. This external reliance introduces a single point of failure (SPOF); any network disruptions or API rate limits could temporarily degrade the system’s conversational fluency, forcing it to rely exclusively on the fallback dataset.

4.4 Future Work

Addressing these limitations forms the foundation of our future developmental roadmap. The immediate next phase of this research will focus heavily on linguistic localization, specifically compiling, annotating, and integrating a specialized Darija/Arabizi dataset to maximize the system’s regional relevance and comprehension. Beyond technical enhancements, we intend to conduct rigorous qualitative evaluations involving licensed Moroccan mental health professionals. These clinical trials will systematically assess the therapeutic appropriateness, cultural sensitivity, and overall clinical safety of the LLM-generated responses. Finally, to ensure widespread accessibility and platform stability, future iterations will include the deployment of a dedicated mobile application built in React Native, alongside a database migration from SQLite to a robust PostgreSQL environment to handle concurrent user scaling.

5 Conclusion

This paper presented the comprehensive architecture and empirical validation of BMA, a hybrid AI-driven conversational agent specifically engineered to provide preliminary men-

tal health support in the Moroccan context. By successfully integrating two computationally efficient, fine-tuned DistilBERT classification models with the dynamic generation capabilities of the LLAMA 3.3 Large Language Model, the system achieved exceptionally high test accuracies—recording 93.8% for intent mapping and 91.2% for emotion detection. Furthermore, the dual-layer architecture maintained an overall operational latency of less than one second, proving its technical viability for seamless, real-time user interaction. A critical component of this system’s success is the implementation of a strict 70% algorithmic confidence threshold alongside a multilingual safety scanner. This mechanism effectively regulates the generative layer, ensuring that users receive reliable, clinically safe interventions and immediate redirections to emergency services during potential crisis scenarios. While the current framework conclusively demonstrates the technical feasibility of a scalable, low-cost psychological assistance tool, it also highlights clear pathways for necessary future development. Subsequent iterations must prioritize the incorporation of localized linguistic nuances, particularly the Moroccan Darija dialect and Arabizi script, to maximize true grassroots accessibility. Ultimately, BMA establishes a secure, robust, and multilingual foundation for deploying continuous mental health resources in regions facing critical shortages of medical professionals, successfully bridging a vital gap in modern digital healthcare.

References

- [1] World Health Organization, “Comprehensive mental health action plan 2013–2030,” WHO, Geneva, Switzerland, 2021. [Online]. Available: <https://www.who.int/publications/i/item/9789240031029>
- [2] CESE (Conseil Économique, Social et Environnemental), “Santé mentale et prévention du suicide : des constats alarmants et des recommandations urgentes,” *CESE*, Rabat, Maroc, Rapport d’auto-saisine, 2021. [Online]. Available: <https://www.cese.ma>
- [3] K. K. Fitzpatrick, A. Darcy, and M. Vierhile, “Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial,” *JMIR Mental Health*, vol. 4, no. 2, e19, 2017. [Online]. Available: <https://mental.jmir.org/2017/2/e19/>
- [4] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” *arXiv preprint arXiv:1910.01108*, 2019. [Online]. Available: <https://arxiv.org/abs/1910.01108>
- [5] Meta AI, “Llama 3 Model Card,” Meta, 2024. [Online]. Available: <https://ai.meta.com/llama/>

- [6] I. Loshchilov and F. Hutter, “Decoupled Weight Decay Regularization,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, New Orleans, LA, USA, 2019. [Online]. Available: <https://arxiv.org/abs/1711.05101>
- [7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in *Proceedings of NAACL-HLT*, Minneapolis, MN, USA, pp. 4171–4186, 2019. [Online]. Available: <https://arxiv.org/abs/1810.04805>
- [8] T. Wolf et al., “Transformers: State-of-the-Art Natural Language Processing,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pp. 38–45, 2020. [Online]. Available: <https://aclanthology.org/2020.emnlp-demos.6/>